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Objectives

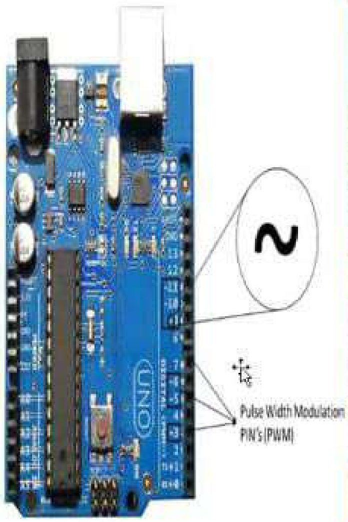
Define the Pulse Width Modulation (PWM)

Control the Pulse Width Modulation

Use the PWM with electric elements

Pulse Width Modulation (PWM)

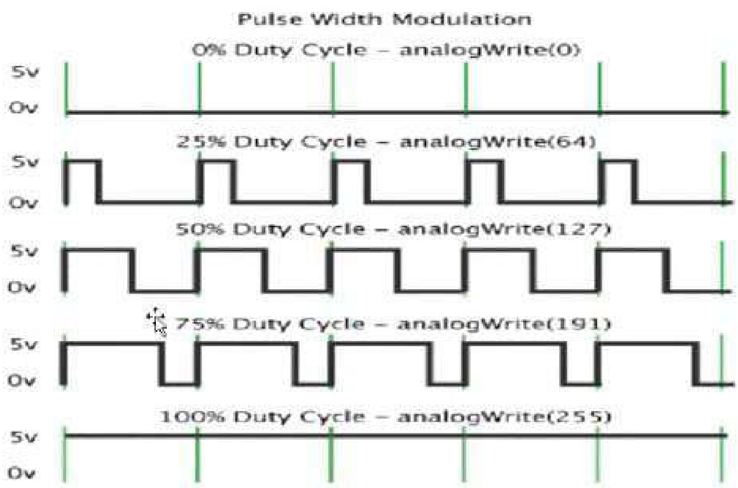
- Pulse Width Modulation, or PWM, is a technique for getting analog results with digital means.
- Digital control is used to create a square wave, a signal switched between on and off. This on-off pattern can simulate voltages in between full on (5 Volts) and off (0 Volts) by changing the portion of the time the signal spends on versus the time that the signal spends off.
- The duration of "on time" is called the pulse width. To get varying analog values, you change or modulate, that pulse width. If you repeat this on-off pattern fast enough with an LED, for example, the result is as if the signal is a steady voltage between 0 and 5v controlling the brightness of the LED."



dot.

Pulse Width Modulation (PWM)

As PWM is how many square waves you are modulating. We can control them by changing the value of square waves. On the figure below you can see the square waves and how to control them with an analogWrite() command.



Pulse Width Modulation (PWM)

$$\text{Duty Cycle} = \frac{T_{on}}{T_{on}+T_{off}} \times 100$$

$\boldsymbol{V_{out}}$ (voltage received by the LED) = Duty cycle x $\boldsymbol{V_{max}}$ ($\boldsymbol{V_{max}}$ = 5V).

PWM reading value are in the range 0 – 255 ($\mathbf{2^8 = 256}$ ***Arduino is bases on 8 bits microcontrollers, this mean they have 8 bits registers***) and the first value is for storing the length of the reading values.

- analogWrite(0) = 0% duty cycle(0V)
- analogWrite(63 or 64) = 25% duty cycle – Low bright (1.25V)
- analogWrite(126 or 127) = 50% duty cycle – Medium bright (2.5V)
- analogWrite(188 or 191) = 75% duty cycle – High bright (3.75V)
- analogWrite(255) = 100% duty cycle – Full bright (5V)

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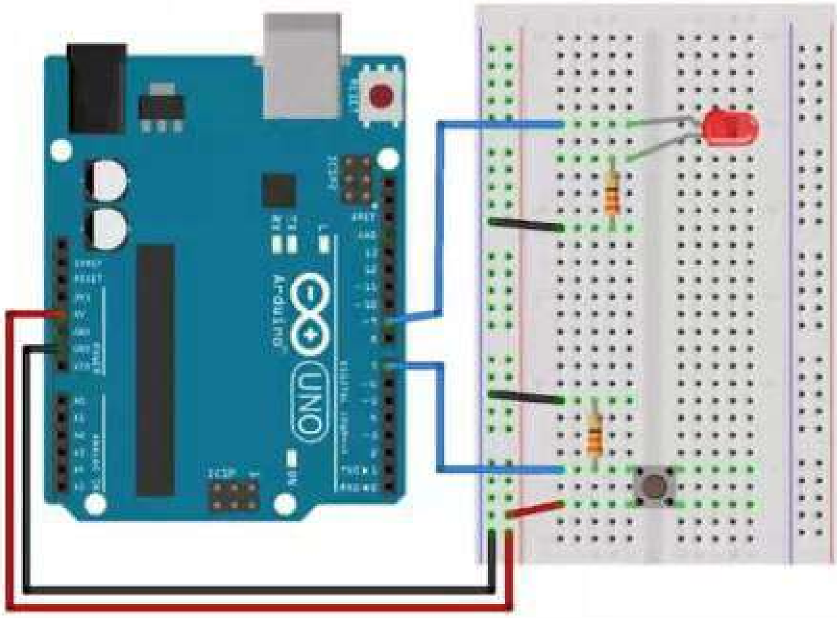
Board connection

Board connection:



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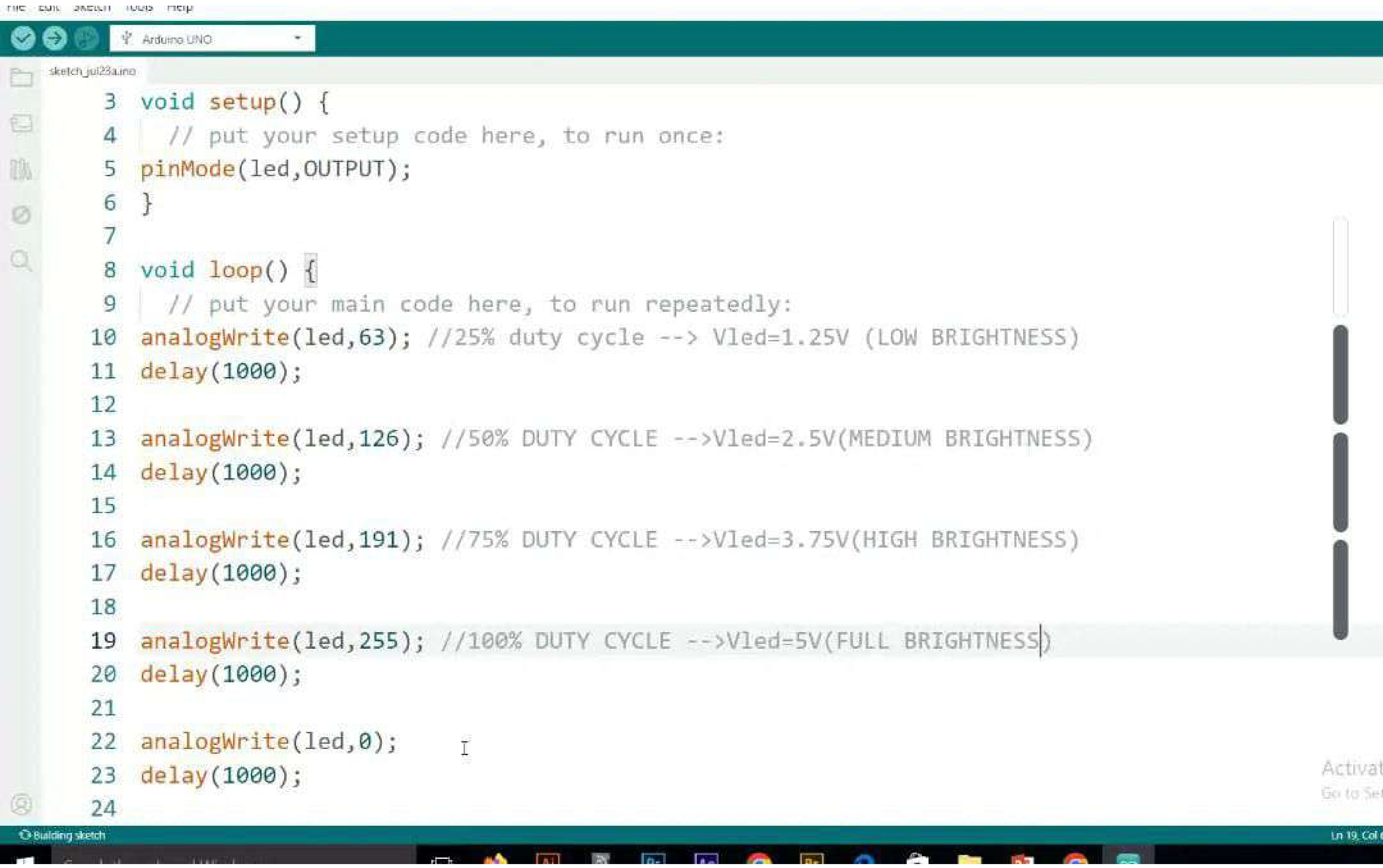
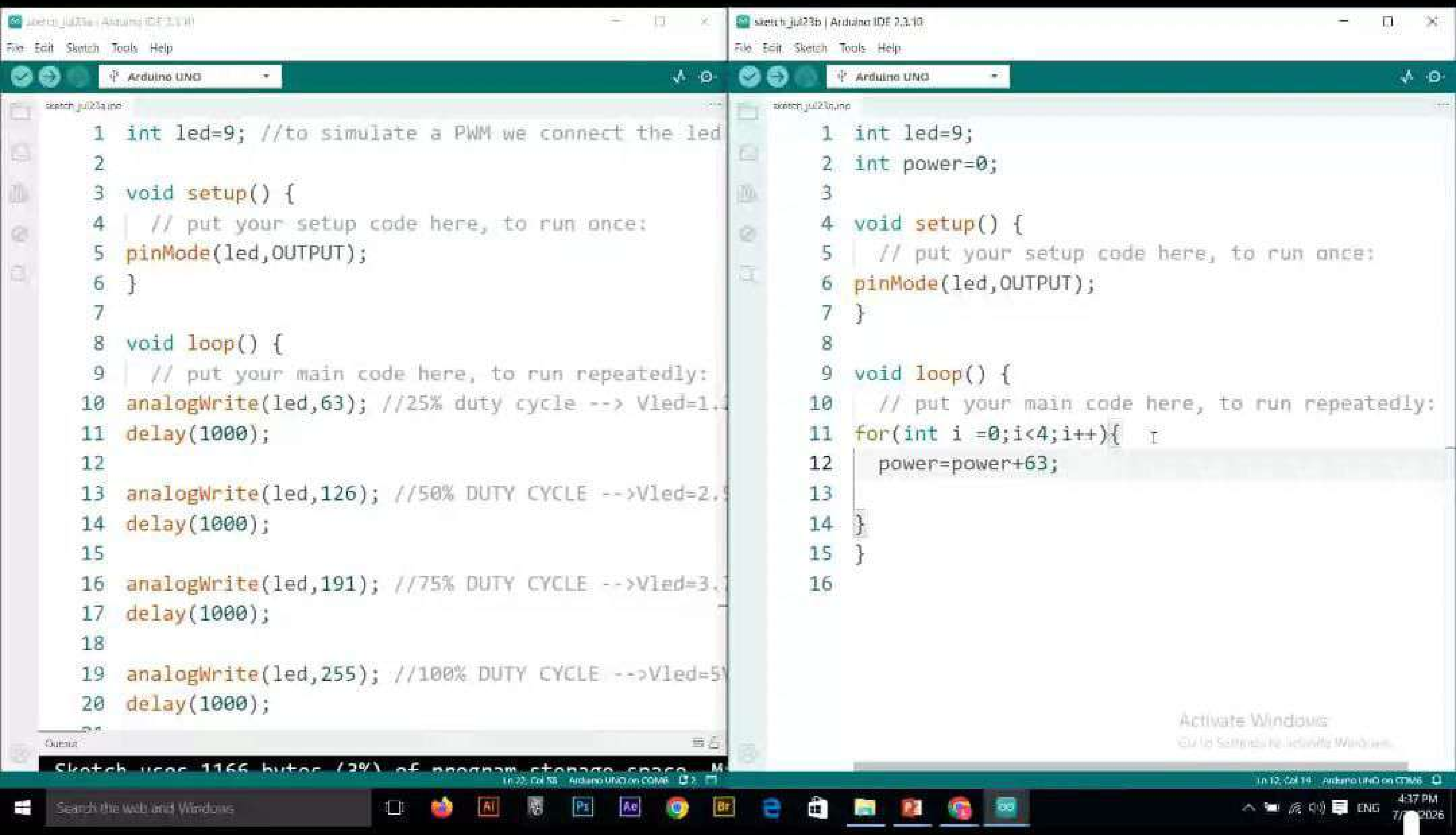
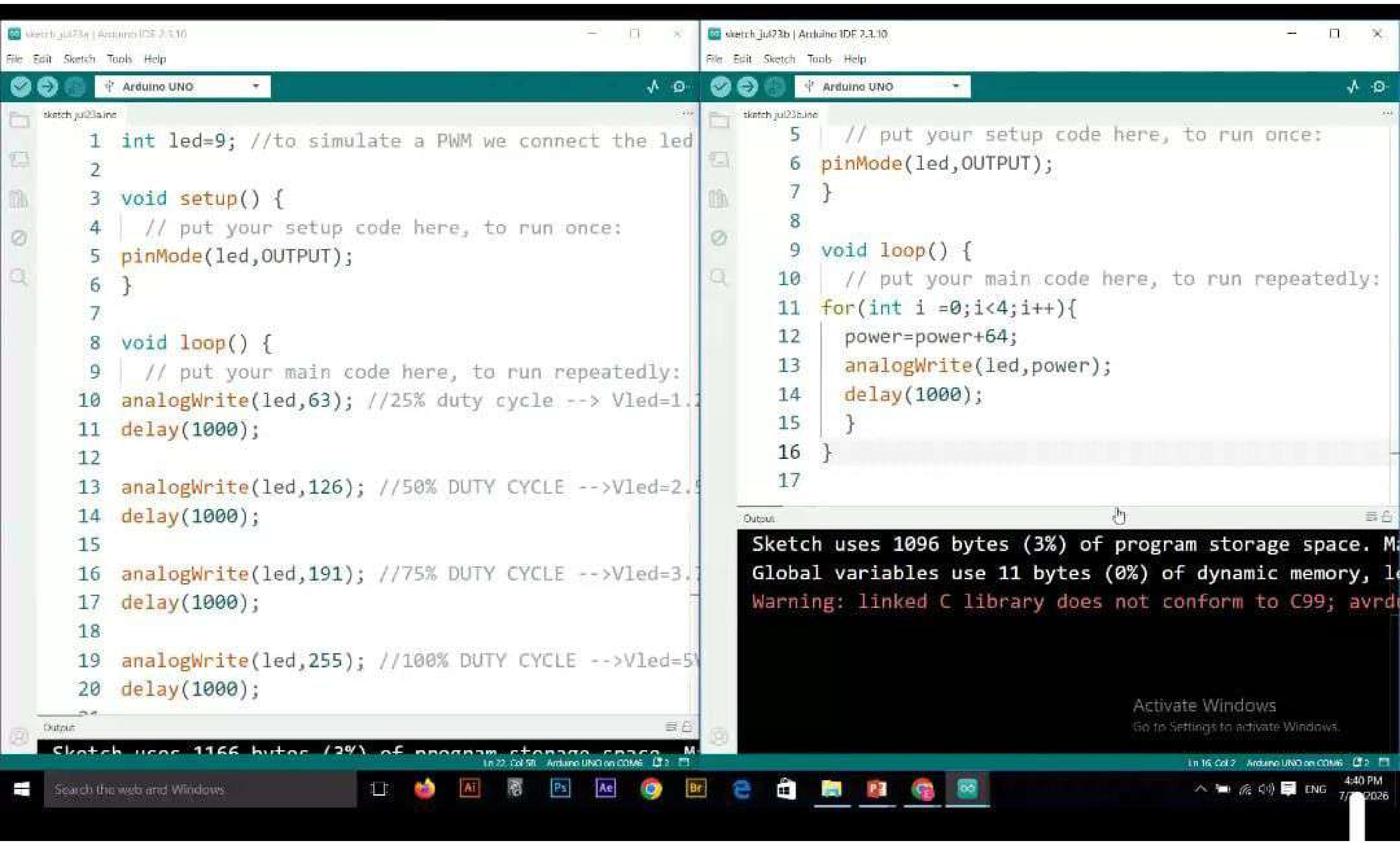
sketch_jul23a.ino

```
void setup() {
  // put your setup code here, to run once:
pinMode(led,OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
analogWrite(led,63);
delay(1000);

analogWrite(led,126);
delay(1000);

analogWrite(led,191);
delay(1000);

analogWrite(led,63);
delay(1000);

analogWrite(led,63);
delay(1000);

```

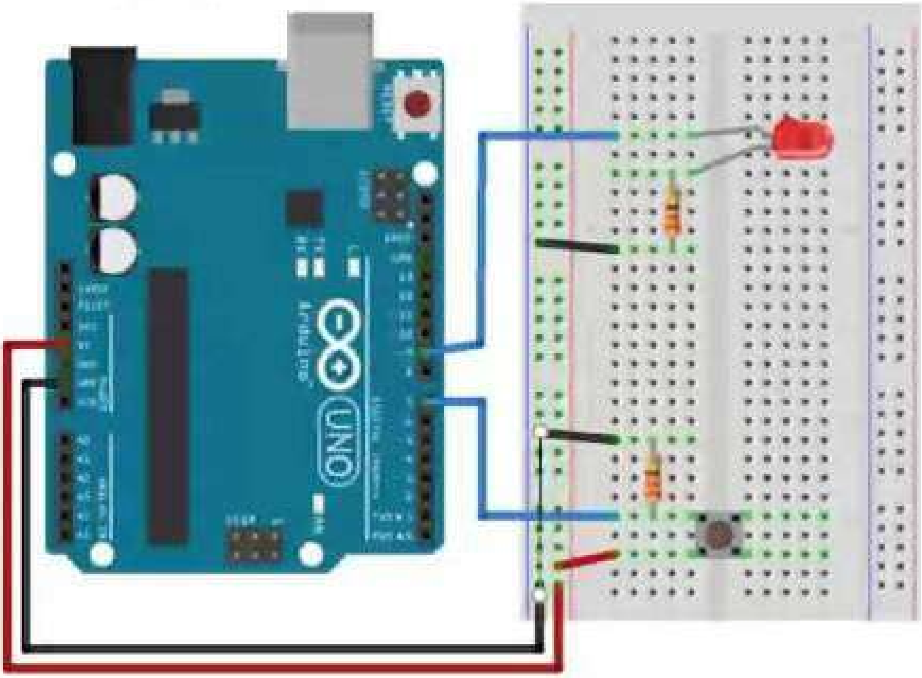

Arduino UNO
sketch_jul23a.ino
void setup() {
// put your setup code here, to run once:
pinMode(led,OUTPUT);
}
void loop() {
// put your main code here, to run repeatedly:
analogWrite(led,63); //25% duty cycle --> Vled=1.25V (LOW BRIGHTNESS)
delay(1000);
analogWrite(led,126); //50% DUTY CYCLE -->Vled=2.5V(MEDIUM BRIGHTNESS)
delay(1000);
analogWrite(led,191); //75% DUTY CYCLE -->Vled=3.75V(HIGH BRIGHTNESS)
delay(1000);
analogWrite(led,255); //100% DUTY CYCLE -->Vled=5V(FULL BRIGHTNESS)
delay(1000);
analogWrite(led,0);
delay(1000);
Building sketch


sketch_jul23b | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
int led=9; //to simulate a PWM we connect the led
void setup() {
// put your setup code here, to run once:
pinMode(led,OUTPUT);
}
void loop() {
// put your main code here, to run repeatedly:
analogWrite(led,63); //25% duty cycle --> Vled=1.
delay(1000);
analogWrite(led,126); //50% DUTY CYCLE -->Vled=2.
delay(1000);
analogWrite(led,191); //75% DUTY CYCLE -->Vled=3.
delay(1000);
analogWrite(led,255); //100% DUTY CYCLE -->Vled=5V
delay(1000);
Sketch uses 1166 bytes (3%) of program storage space. M
int led=9;
int power=0;
void setup() {
// put your setup code here, to run once:
pinMode(led,OUTPUT);
}
void loop() {
// put your main code here, to run repeatedly:
for(int i =0;i<4;i++){
power=power+63;
}
}
Activate Windows
Search the web and Windows
ENG
4:37 PM


sketch_jul23b | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
sketch_jul23a.ino
int led=9; //to simulate a PWM we connect the led
void setup() {
// put your setup code here, to run once:
pinMode(led,OUTPUT);
}
void loop() {
// put your main code here, to run repeatedly:
analogWrite(led,63); //25% duty cycle --> Vled=1.
delay(1000);
analogWrite(led,126); //50% DUTY CYCLE -->Vled=2.
delay(1000);
analogWrite(led,191); //75% DUTY CYCLE -->Vled=3.
delay(1000);
analogWrite(led,255); //100% DUTY CYCLE -->Vled=5
delay(1000);
Output
Arduino UNO
sketch_jul23b.ino
// put your setup code here, to run once:
pinMode(led,OUTPUT);
}
void loop() {
// put your main code here, to run repeatedly:
for(int i =0;i<4;i++){
power=power+64;
analogWrite(led,power);
delay(1000);
}
}
Output
Sketch uses 1096 bytes (3%) of program storage space.
Global variables use 11 bytes (0%) of dynamic memory,
Warning: linked C library does not conform to C99;
Activate Windows
Go to Settings to activate Windows.
Search the web and Windows
ENG
4:40 PM

Board connection

Board connection:



dot.